	PC	Fundamentals of Signal Processing & Communication	3	1	0	4	4
	PC	Computer Organization & Microprocessor	3	0	0	3	3
	PC	Computer Organization & Microprocessor Laboratory	0	0	2	2	1
			17	**1**	**8**	**26**	**22**
Sem V	OE	Open Elective 2 **((NPTEL/SWAYAM/MOOC))**	3	0	0	3	3
	HSC	Engineering Economics	3	0	0	3	3
	PE	Program Elective 1	3	0	0	3	3
	PC	Theory of Computation & Compiler Design	3	0	0	3	3
	PC	Theory of Computation & Compiler Design Laboratory	0	0	2	2	1
	PC	Digital Signal Processing	3	0	0	3	3
	PC	Digital Signal Processing Laboratory	0	0	2	2	1
	PC	RF Engineering	3	0	0	3	3
	PC	RF Engineering Laboratory	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VI	OE	Open Elective 3 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 2	3	0	0	3	3
	PE	Program Elective 3	3	0	0	3	3
	PC	Operating Systems	3	0	0	3	3
	PC	Operating Systems Laboratory	0	0	2	2	1
	PC	Digital Communication	3	0	0	3	3
	PC	Digital Communication Laboratory	0	0	2	2	1
	PC	Computer Communication and Networking	3	0	0	3	3
	PC	Computer Communication and Networking Laboratory	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VII	Pro	Summer Internship	0	0	0	0	2
	OE	Open Elective 4 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 4 (3-0-0)	3	0	0	3	3
	PE	Program Elective 5 (3-0-0)	3	0	0	3	3
	PC	Software Engineering Methodology	3	0	0	3	3
	PC	Software Engineering Methodology Laboratory	0	0	2	2	1
	PC	Embedded Systems	3	0	0	3	3
	PC	Embedded Systems Laboratory	0	0	2	2	1
	PC	Digital CMOS and VLSI Design	3	0	0	3	3
	PC	Digital CMOS and VLSI Design Laboratory	0	0	2	2	1
		Seminar					1
			18	0	6	24	24
Sem VIII		**Course Name**	**Th**	**Tut**	**Pra**	**Hrs**	**Cr**
	Pro	Major/Comprehensive Project					**12**

Program Core Electives:

Semester	Software and Networks	Artificial Intelligence	Communication and Signal Processing	Embedded and VLSI
5 (program core elective1)	Web Technology	Optimization methods and Algorithms		Introduction to CMOS and Memory Technology
	Problem solving through Java			
6 (program core elective2, 3)	Cloud Architecture and Service	Machine Learning	Image Processing	Hardware Accelerated Computing
	Advanced Web Technology	Data warehousing & Mining	Statistical Signal Processing	
	Advanced Algorithm Design			
7 (program core elective4, 5)	Cryptography and Network Security	Computer Vision	Optical Communication	Autonomous systems
	Internet of Things	Deep Learning and Reinforcement learning	Modern Wireless Communications	Hardware Accelerated Computing
	Blockchain Technology			Real time Operating Systems
	Mobile Application Development			

5th Semester

<Course Code>					Theory of Computation and Compiler Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. Provides the required theoretical foundation for a computational model and compiler design
2. Discuss various types of machines as an abstract computational model
3. Study and design various methods to develop different phases of compiler

UNIT 1 LANGUAGES, GRAMMARS AND AUTOMATA	**12 Hrs.**
Fundamentals of formal languages, grammars, and computation models, Introduction to compilers and its Phases, Finite Automata, Regular Language and Regular Expressions, Deterministic and Non-deterministic Finite Automata, Minimization of DFA, Conversion of RE to DFA using Arden's Formula, Pumping Lemma, Design Lexical Analyser using Finite Automata	
UNIT 2 CFG, CFL, PDA, and PARSERS	**12 Hrs.**
Push Down Automata (PDA), Deterministic and Non-deterministic PDA, Context Free Grammar (CFG), Left Most Derivation, Right Most Derivation, Parse Tress, Ambiguity in languages, Normal Forms, Context Free Language (CFL), Pumping Lemma for CFL, CYK membership algorithms, Syntax analysis, Top-Down parsing, Bottom-Up parsing, Operator-Precedence parsing	
UNIT 3 TURING MACHINES AND INTERMEDIATE CODE GENERATION PHASE	**9 Hrs.**
Turing Machines, Recursive and recursively enumerable languages, Chomsky's hierarchy, Halting problem, Universal Turing Machine, The Classes P and NP, An NP-complete problem, Intermediate Code Generation, Intermediate Languages, Declarations, Assignment Statements, Boolean Expressions, Case Statements, etc.	
UNIT 4 CODE OPTIMIZATION AND CODE GENERATION	**9 Hrs.**
Code optimization phase of the compiler, Basic blocks and flow graphs, The DAG representation, The principal sources of optimization, code optimization techniques, Code generation techniques, Issues in the design of a code generator	

Max. 42 Hrs.

COURSE OUTCOMES

On completion of the course, students will be able to

CO1	:	Remember the concepts of formal languages, grammar and automata
CO2	:	Comprehend various computational models
CO3	:	Apply the knowledge of theory of computation and automata to study phases of compilers.
CO4	:	Analyse the different types of parsers.
CO5	:	Evaluate different compiler design techniques
CO6	:	Design all the seven phases of compiler for a specific language

TEXT/REFERENCE BOOKS

1. Introduction to Automata Theory, Languages, and Computation (3rd Edition), John E Hopcroft, Rajeev Motwani, Jeffery D. Ullman, Pearson Education, 2013.
2. Compilers: Principles, Techniques, and Tools, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, 2nd Edition, Pearson Education

<Course Code>					Theory of Computation and Compiler Design Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

1. Provides the required theoretical foundation for a computational model and compiler design
2. Discuss various types of machines as an abstract computational model
3. Study and design various methods to develop different phases of compiler

The lab experiment could be based on following topics:

1. Understanding concepts of Language, Alphabet, Operation on Language
2. Design and write a program for DFA
3. Study of DFA, NDF and conversions
4. Minimization of DFA
5. Study different phases of compiler and design Lexical analyser by token passing
6. Understanding CFG and CFL
7. Designing the PDA
8. Designing the Top-down parser
9. Designing the Bottom-up parser
10. Study and design of Turing Machine
11. Study of code synthesis phases of compiler

COURSE OUTCOMES

On completion of the course, students will be able to

CO1	:	Remember the concepts of formal languages, grammar and automata
CO2	:	Comprehend various computational models
CO3	:	Apply the knowledge of theory of computation and automata to study phases of compilers.
CO4	:	Analyse the different types of parsers.
CO5	:	Evaluate different compiler design techniques
CO6	:	Design all the seven phases of compiler for a specific language

TEXT/REFERENCE BOOKS

1. Introduction to Automata Theory, Languages, and Computation (3rd Edition), John E Hopcroft, Rajeev Motwani, Jeffery D. Ullman, Pearson Education, 2013.
2. Compilers: Principles, Techniques, and Tools, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, 2nd Edition, Pearson Education

24ICxxxT					Digital Signal Processing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Introduce mathematical tools for analysis of continuous time signals and LTI systems.
- Understand basic principles of communication systems.
- Create foundation for digital signal processing.

UNIT 1: ANALYSIS OF DISCRETE TIME SIGNALS AND SYSTEMS	10 Hrs.
Introduction to DSP, A review of Sampling Theorem, Discrete Time Signals and Systems, Z-transform, ROC and Properties, Poles- Zeros, Inverse z-transform, LTI System Analysis using Z-transform, Discrete Time Fourier Transform (DTFT) and Important Properties, Frequency Response of Discrete Time Systems, Correlation of Discrete Time Signals, Parseval's relation, Energy and Power Spectral Density.	
UNIT 2: DIGITAL FILTERS: DESIGN AND STRUCTURES	**11 Hrs.**
Ideal Digital Filters, Practical Filters: Stability and Causality, FIR and IIR Filters, Linear Phase and Implications, Filter Design Steps, Design of Linear Phase FIR Filters, Window Method, IIR Filter Design, Pole-Zero Placement Method, Analog Filter Design, Analog Filter Standard Frequency Responses and Design Equations, IIR Filter Coefficients from Analog Filter, Bilinear Transformation, Basic Structures for FIR and IIR Systems implementation, Direct, Transposed Cascade and Parallel form Structures, Effects of Co-efficient Quantization.	
UNIT 3: DFT AND FFT ALGORITHMS	**11 Hrs.**
Effect of periodicity and discretization on spectra of a signal, Sampling of DTFT, DFT and IDFT, Important Properties of DFT, Linear and Circular Convolution, Efficient Computation of DFT, Radix2 FFT Algorithms, Geortzel Algorithm.	
UNIT 4: FOURIER ANALYSIS OF SIGNAL AND SYSTEM USING DFT AND HILBERT TRANSFORM	**10 Hrs.**
DFT Analysis of Sinusoidal Signals; The Time-Dependent Fourier Transform; Block Convolution using the Time Dependent Fourier Transform; Application of DFT in Linear Filtering, Fourier Analysis of Nonstationary Signals; Fourier Analysis of Stationary Random Signals: The Periodogram, Fourier Transform for Causal Sequences; Relationships between Magnitude and Phase; Hilbert Transform Relations for Complex Sequences	

Total: 42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Remember properties of signals, systems, and principles of various modulation and demodulation methods.
CO2: Understand mathematical representation of signals, systems, modulated signals and detection techniques.
CO3: Apply mathematical concepts to model behavior of LTI systems, modulation and demodulation of signals.
CO4: Analyze signals and systems in time and frequency domain to obtain various parameters.
CO5: Evaluate properties of LTI systems and performance of communication systems.
CO6: Design a block diagram level elementary communication and signal processing system

TEXT/REFERENCE BOOKS

4. Alan V.Oppenheim, Alan S.Willsky with S. Hamid Nawab, Signals & Systems, Pearson Education.
5. B. P. Lathi, Signal Processing and Linear System, Berkeley Cambridge Press.
6. B.P.Lathi, Zhi Ding, Modern Digital and Analog Communication Systems, Oxford University Press.
7. P. Chakrabarti, Analog and Digital Communication, Dhanpat Rai & Co.
8. J. G.Proakis, D. G.Manolakis, Digital Signal Processing, Principles, Algorithms and Applications, PHI.

24ICxxxP					Digital Signal Processing Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	25	25	50

COURSE OBJECTIVES

- To impart knowledge of flow process of digital signal processing.
- To implement digital signal processing algorithms on software platforms.
- To evaluate performance of digital signal processing systems and algorithms

Laboratory Sessions would be based on following topics but not limited to:

1. Discrete time signal generation and applying different operations on signals.
2. Z-domain and frequency domain analysis of discrete time signals and systems.
3. Design and implementations of FIR and IIR filters.
4. Implementation and application of DFT-FFT in linear filtering.
5. Simulation of discrete time systems for speech and audio applications.
6. Implementation of digital signal processing algorithm.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Remember representation of discrete time signals and systems in Z-domain and frequency domain.
CO2: Understand the methodologies to utilize digital signal processing techniques practically.
CO3: Apply concepts to realize discrete time systems and digital signal processing algorithms practically.
CO4: Analyze and report the behaviour of discrete time systems and digital signal processing algorithms.
CO5: Evaluate and report performance of discrete time systems and digital signal processing algorithms.
CO6: Design and implement practical applications using software development tools.

TEXT/REFERENCE BOOKS

1. 1.Alan V.Oppenheim, Alan S.Willsky with S. Hamid Nawab, Signals & Systems, Pearson Education.
2. B. P. Lathi, Signal Processing and Linear System, Berkeley Cambridge Press.
3. B.P.Lathi, Zhi Ding, Modern Digital and Analog Communication Systems, Oxford University Press.
4. P. Chakrabarti, Analog and Digital Communication, Dhanpat Rai & Co.
5. J. G.Proakis, D. G.Manolakis, Digital Signal Processing, Principles, Algorithms and Applications, PHI.

24ICxxxT					RF Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To Impart electromagnetic and transmission line concepts
- To learn the radiating body behavior for wireless communication system
- To introduce the wave propagation mechanism in different modes
- To elaborate and familiarize the different modern RF technologies

UNIT Electromagnetics:	**17Hrs.**
Maxwell's equations, pointing theorem, uniform plane wave and reflection, plane wave propagation in free space and lossless & lossy media. Transmission line: Types of transmission lines, transmission line equation and solution, reflection and transmission coefficients, standing wave and standing wave ratio, line impedance, Smith chart, impedance matching. Waveguides, waveguide Tees, bands, corners, twists, posts, irises, directional couplers, Faraday's rotation, circulator and isolator, Introduction, modes (TEM, TEmn, TMmn) and field pattern in medium, S-parameters.	
UNIT 2: RADIATING SYSTEMS	**10 Hrs.**
Basics: Radiation Mechanism and Current Distribution, Fundamental Parameters related to antenna (Radiation Pattern, Radiation Power Density, Directivity, Gain, Beam width, Bandwidth, Polarization, Radiation Efficiency, Antenna Efficiency, Far Field Radiation. Antenna Array: Two Element Array N-Element Linear Array with uniform amplitude and spacing, Broadside and End-Fire Arrays, N-Element 2D Array. Microstrip antenna: Introduction to strip line and microstrip lines, Advantages and limitations, feeding techniques, introduction to rectangular microstrip antenna, RMSA and CMSA design, filters design using microstrip line.	
UNIT 3 : RADAR AND WAVE PROPAGATION	**8 Hrs.**
Introduction (basic principle and types) to radar, radar range equation, maximum unambiguous range, pulse & MTI radars blind speed, radar resolution. Modes of wave propagation, Ground Wave Propagation, Sky wave propagation, Definitions for Virtual height, MUF and Skip distance, Multi-hop propagation, Space wave propagation..	
UNIT 4 : MODERN RF TECHNOLOGIES	**7 Hrs.**
RFID, NFC technology and their interfacing with IoT, RF Energy harvesting for WSN & IoT, 5G Technology, Antenna Design for 5G, MU-MIMO and Massive MIMO for5G Radios, study of effect of RF on flora fauna and human being.	

Total: 42Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Remembering the fundamentals of electromagnetics, radiating structures, wave propagation and modern RF techniques.

CO2: Understanding effect and outcome of RF waves propagating in different mediums including transmission line, free space and the under influence of RF Techniques.

CO3: Applying the concepts of transmission medium, radiating elements and wave propagation in developing new systems.

CO4: Analyzing the behaviour of wave and its impact on passive component properties in different transmission conditions

CO5: Evaluating different RF Technologies, components and mediums for RF Waves.

CO6: Design RF subsystems utilizing the radiating elements, passive components utilizing different mediums for developing different technologies.

TEXT/REFERENCE BOOKS

1. Samuel Y. Liao, "Microwave Devices and Circuits", Pearson Education, 3rd edition, 2012.
2. C. A. Balanis, "Antenna theory: analysis and design", John Wiley and sons, 3rd Edition, 2005.
3. M. L. Skolnik, "Introduction to Radar Systems", Mc Graw Hill, 3rd edition, 2016.
4. K. D. Prasad, "Antenna and wave Propagation", Satya Prakashan, 3rdedition, 2009.

24ICTxxxP					RF Engineering Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	0	0	0	50	50	100

COURSE OBJECTIVES:

- To Impart electromagnetic and transmission line concepts
- To learn the radiating body behavior for wireless communication system
- To introduce the wave propagation mechanism in different modes

To elaborate and familiarize the different modern RF technologies

List of Experiments: Following is the list of experiments of experiments but it is not limited to it.

1. To measure position of fault in cable
2. To measure the frequency and wavelength using slotted line section and frequency meter.
3. To measure the Isolation and Insertion loss of Isolator and Circulator.
4. To study E-plane/H-plane/Magic Tee.
5. To measure Coupling Factor, Directivity and Isolation of directional coupler.
6. To measure VSWR and Reflection coefficient of different loads.
7. To study the characteristics of Klystron and Gunn diode.
8. To study the behavior of Transmission line: Waveguide/Coaxial cable/microstrip line.*
9. To study the S-parameter of directional coupler.*
10. To study the S-parameter of E-plane/H-plane Tee.*
11. To design and study MIC Based LPF.*

*** Some of these experiments need to be using simulation software.**

COURSE OUTCOMES:

On completion of the course, student will be able to

CO1: Remembering the fundamentals of electromagnetics, radiating structures, wag=ve propagation and modern RF techniques.
CO2: Understanding effect and outcome of RF waves propagating in different mediums including transmission line, free space and the under influence of RF Techniques.
CO3: Applying the concepts of transmission medium, radiating elements and wave propagation in developing new systems.
CO4: Analyzing the behaviour of wave and its impact on passive component properties in different transmission conditions
CO5: Evaluating different RF Technologies, components and mediums for RF Waves.
CO6: Design RF subsystems utilizing the radiating elements, passive components utilizing different mediums for developing different technologies.

TEXT/REFERENCE BOOKS

9. Samuel Y. Liao, “Microwave Devices and Circuits”, Pearson Education, 3rd edition, 2012.
10. C. A. Balanis, “Antenna theory: analysis and design”, John Wiley and sons, 3rd Edition, 2005.
11. M. L. Skolnik, “Introduction to Radar Systems”, Mc Graw Hill, 3rd edition, 2016.
12. K. D. Prasad, “Antenna and wave Propagation”, Satya Prakashan, 3rdedition, 2009.

Program Core Elective

5th Semester

<Course Code>					Optimization Methods and Algorithms					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To learn the fundamentals of optimization techniques.
2. To be able to understand, analyze, and solve constrained and unconstrained optimization problems.
3. To apply optimization techniques to solve diverse problems.

UNIT I: INTRODUCTION AND MATHEMATICAL PRELIMINARIES	**09 Hrs.**
Introduction to optimization and engineering optimization problems, Classification of optimization problems and algorithms, Simple application examples, Univariate and multi variate general (nonlinear) optimization problem formulations: Maxima and Minima. Mathematical Preliminaries: vectors, norm, inner product, matrices, eigen values and vectors, Definiteness, Multi-variate linear, affine and quadratic functions, Gradient vector and Hessian matrix, Matrix calculus, Convex/Concave set and functions.	
UNIT II: MULTIVARIATE OPTIMIZATION METHODS AND ALGORITHMS	**15 Hrs.**
Multi-variate non-linear optimization methods without constraints: Optimality conditions, Gradient based optimization algorithms: Newton's method, steepest descent, line search, conjugate gradient., Gradient free and approximation algorithms: Downhill Simplex, BFGS and Trust region algorithms., Equality constraints: Lagrange Multiplier method., Inequality constraints: KKT conditions, Convex optimization methods: Quadratic programming, Least square problem, Penalty function, Minimax optimization, Solution using KKT conditions, Newton's method, Barrier functions and interior point methods, Sequential quadratic programming, Separable programming.	
UNIT III: LINEAR AND INTEGER PROGRAMMING METHODS AND ALGORITHMS	**9 Hrs.**
Linear programming: Graphical method, Simplex algorithm, Duality and applications, Integer linear programming, Cutting plane and Branch and bound methods, Applications.	
UNIT IV: MODERN METHODS OF OPTIMIZATION – META-HEURISTIC TECHNIQUES	**9 Hrs.**
Introduction to nature inspired optimization, Genetic Algorithms, Simulated Annealing, Particle Swarm Optimization, Ant Colony Optimization, Applications, Introduction to multi-objective optimization.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Understand the need of optimization methods for engineering problems.
CO2	:	Describe unconstrained optimization and constrained optimization methods and algorithms.
CO3	:	Apply optimization techniques to solve linear and non-linear optimization problems.
CO4	:	Analyze and formulate different optimization problems.
CO5	:	Evaluate performance of various optimization techniques and algorithms.
CO6	:	Develop solution to various engineering problems with different level of constraints.

TEXT/REFERENCE BOOKS

5. Xin-She Yang, "Optimization Techniques and Applications with Examples", John Wiley & Sons Publication.
6. Xin-She Yang, "Engineering Optimization: An Introduction with Metaheuristic Applications", John Wiley & Sons Publication.
7. S.S. Rao, "Engineering Optimization: Theory and Practice", John Wiley & Sons Publication.
8. Kalyanmoy Deb, "Optimization For Engineering Design: Algorithms and Examples", Prentice Hall India Learning Private Limited, 2012.

24IC301T					Web Technology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	1	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Learn fundamentals of web development.
- To learn client side scripting with JavaScript.
- Demonstrate the use of ECMAScript to fulfil the essentials of front-end development.
- Implement UI-UX using React.js library and to create responsive web applications.

UNIT 1 Foundations of Web Development and Design	**7 Hrs.**
Introduction to WWW : Protocols and programs, secure connections, application and development tools, the web browser, What is server, choices, setting up UNIX and Linux web servers, Logging users, dynamic IP Web Design: Web site design principles, planning the site and navigation, Introduction to HTML : The development process, Html tags and simple HTML forms, web site structure Introduction to XHTML : XML, Move to XHTML, Meta tags, Character entities, frames and frame sets, inside browser	
UNIT 2: Styling and Interactivity with CSS and JavaScript	**6 Hrs.**
Style sheets : Need for CSS, introduction to CSS, basic syntax and structure, using CSS, background images, colors and properties, manipulating texts, using fonts, borders and boxes, margins, padding lists, positioning using CSS, JavaScript : Client side scripting, What is JavaScript, How to develop JavaScript, simple JavaScript, variables, functions, conditions, loops and repetition	
UNIT 3 : Modern JavaScript and ECMAScript	**7 Hrs.**
Modern JavaScript concepts, JavaScript and objects, JavaScript own objects, the DOM and web browser environments, forms and validations DHTML : Combining HTML, CSS and JavaScript, events and buttons, controlling your browser, Introduction to ECMAScript 6 JavaScript Variables - var, let, const, JavaScript Objects, JavaScript Events, JavaScript Array Methods, JavaScript Regular Expressions DOM Manipulation JavaScript Best Practices	
UNIT 4 : Front-end Web Development with React.js	**8 Hrs.**
Introduction to Frontend web development, Single page web applications, Fundamentals of React.js, Components: Class Components and Functional Components, React Hooks - useEffect, useReducer, useContext, useCallback, fetching data with React Hooks, JSX, Props, Methods as props, State and setState, Destructuring props and state.	

Total: 280rs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Learn the Web Design Concepts including WWW, HTTP protocol and Browser.
CO2 – Understand the design and style concepts of webpages using HTML and CSS
CO3 – Implement JavaScript functionality to make interactive webpages
CO4 – Demonstrate the use of ECMAScript 6 to fulfil the essentials of front-end development.
CO5 – Implement UI-UX using React.js framework and Responsive web designing for Web Applications
CO6 – Build a complete web solution for a given problem statement

TEXT/REFERENCE BOOKS

1. Terry Felke-Morris, Web Development and Design Foundations with HTML5 - Pearson Education.
2. David Flanagan, JavaScript: The Definitive Guide– O'REILLY .
   David Griffiths and Dawn Griffiths, React Cookbook – O'REILLY

<Course Code>					Problem solving through Java					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the basic concepts and fundamentals of platform independent object-oriented language.
2. To demonstrate skills in writing programs using exception handling techniques and multithreading.
3. To understand streams and efficient user interface design techniques

UNIT I: CONCEPTS OF CLASSES AND OBJECTS	**12 Hrs.**
Introduction to java, java buzzword, data types, dynamic initialization, scope and life time, operators, control statements, arrays, type conversion and casting, finals & blank finals. Classes and Objects: Concepts, methods, constructors, usage of static, access control, this key word, garbage collection, overloading, parameter passing mechanisms, nested classes and inner classes. Basic concepts, access specifiers, usage of super key word, method overriding, final methods and classes, abstract classes, dynamic method dispatch, Object class	
UNIT II: INTERFACES	**10 Hrs.**
Differences between classes and interfaces, defining an interface, implementing interface, variables in interface and extending interfaces. Packages: Creating a Package, setting classpath, Access control protection, importing packages. Exception Handling: Concepts of Exception handling, types of exceptions, usage of try, catch, throw, throws and finally keywords, Built-in exceptions, creating own exception sub classes.	
UNIT III: STRINGS	**10 Hrs.**
Exploring the String class, String buffer class, Command-line arguments. Library: Date class, Wrapper classes Multithreading: Concepts of Multithreading, differences between process and thread, thread life cycle, Thread class, Runnable interface, creating multiple threads, Synchronization, thread priorities, inter thread communication, daemon threads, deadlocks.	
UNIT IV: APPLETS AND EVENT HANDLING	**10 Hrs.**
Concepts of Applets, life cycle of an applet, creating applets, passing parameters to applets, accessing remote applet, Color class and Graphics. Events, Event sources, Event classes, Event Listeners, Delegation event model, handling events.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Understand the use of abstract classes.
CO2	:	Understand the reusable programs using the concepts of inheritance, polymorphism, interfaces, and packages.
CO3	:	Apply the concepts of Multithreading and Exception handling to develop efficient and error free codes.
CO4	:	Utilize modern tools and collection framework to create java applications to solve real world problems.
CO5	:	Develop applets for web applications.
CO6	:	Design event driven GUI and web related applications which mimic the real word scenarios.

TEXT/REFERENCE BOOKS

1. Herbert Schildt, "**Java the Complete Reference 9th Edition**", McGraw Hill Education (India) Private Limited, New Delhi.
2. H.M. Dietel and P.J.Dietel, "**Java How to Program**", Sixth Edition, Pearson Education/PHI.
3. Y.DanielLiang "**Introduction to Java programming**", By, Pearson Publication